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1. Title

Make the title specific and academic.

Example:

Development and Evaluation of a Mobile Application for Managing Student Attendance

Avoid titles such as *"My Application Project"* because they don't tell the reader what was investigated.

2. Abstract

Usually around **150–250 words**. Summarise the entire study in one section.

Include:

- **Background/problem** – What problem exists?
- **Objective** – What did you aim to achieve?
- **Method** – How did you develop the application?
- **Results** – What did you find?
- **Conclusion** – What does the study demonstrate?

A simple formula is:

Problem → Objective → Method → Results → Conclusion

Example

This study presents the development of a mobile application designed to improve student attendance management. Conventional attendance recording methods can be time-consuming and may result in inaccurate or incomplete records. The objective of this study was to develop an application that enables lecturers to record, manage, and monitor student attendance digitally. The application was developed using an iterative development approach consisting of requirements analysis, system design, implementation, testing, and evaluation. User interface and database components were integrated to support attendance recording and reporting. The application was evaluated through functional testing and user feedback. The results indicated that the developed application successfully performed the required attendance management functions and improved the efficiency of attendance recording. The findings suggest that the application can provide a practical alternative to conventional attendance management methods.

3. Introduction

The introduction answers:

“Why was this project necessary?”

A good introduction normally moves from broad → specific:

Paragraph 1 — Background

Explain the general situation.

The increasing use of digital technologies has changed the way organisations manage information and deliver services. Applications have become important tools for automating tasks that were previously performed manually.

Paragraph 2 — Problem statement

Explain the specific problem.

However, [target users] still rely on [current method]. This approach can result in [problem 1], [problem 2], and [problem 3].

Paragraph 3 — Existing solutions

Discuss what other researchers/applications have done.

Previous studies have proposed several approaches to address this problem. However, these solutions may have limitations related to [cost/usability/accessibility/features/etc.].

This is where you need **academic references**.

Paragraph 4 — Your solution

Introduce your application.

Therefore, this study proposes the development of [application name], a [web/mobile/desktop] application designed to [main purpose].

Paragraph 5 — Objectives/contribution

State what the research does.

For example:

The objectives of this study are:

1. To identify the requirements for...
2. To design and develop...
3. To evaluate the usability/functionality/performance of...

4. Literature Review

This section answers:

"What has already been done, and what is missing?"

Don't simply describe 10 papers one after another.

Instead, **compare them**.

For example:

Study	Application/Approach	Strength	Limitation
Author A (2023)	Mobile application	Easy to use	Limited reporting
Author B (2024)	Web system	Good data management	Requires internet
Author C (2024)	AI-based system	Automated process	High implementation cost

Then explain the **research gap**:

Although previous studies have developed applications for [problem], limited attention has been given to [specific gap]. Therefore, this study focuses on developing an application that addresses this limitation by...

This is one of the most important parts of a journal paper.

5. Methodology

Explain **how you developed and evaluated the application**.

You can divide it into:

5.1 Development Methodology

For example:

- Agile
- Scrum
- Waterfall
- Prototyping
- Iterative development

Explain why you selected it.

5.2 Requirements Analysis

Describe:

- Target users
- Functional requirements
- Non-functional requirements
- System requirements

For example:

The primary users of the system are students and administrators. Students can register, log in, submit requests, and view their records, while administrators can manage users and generate reports.

5.3 System Design

Include appropriate diagrams:

- Use case diagram
- Activity diagram
- System architecture
- Flowchart
- ERD/database design

5.4 Application Development

Explain your technologies.

For example:

The application was developed using Flutter for the front-end interface and Firebase for data storage and authentication.

Explain **why** these technologies were selected rather than simply listing them.

5.5 Testing and Evaluation

Describe how you determined whether the application worked.

Possible methods include:

- Functional testing
- Usability testing
- User Acceptance Testing (UAT)
- Performance testing
- Questionnaire
- SUS (System Usability Scale)

For example:

A usability evaluation was conducted with 30 participants. Participants were asked to perform five tasks using the application and subsequently complete a questionnaire measuring ease of use, satisfaction, and usefulness.

6. Results

This section reports **what you actually found**.

Don't just say:

"The application works successfully."

Give evidence.

For example:

The functional testing consisted of 25 test cases covering user registration, authentication, data submission, record management, and report generation. Of the 25 test cases, 24 passed during the first testing cycle. The remaining issue involved incorrect validation of an empty input field. After the validation logic was modified, all 25 test cases passed.

You can use:

- Tables

- Figures
- Screenshots
- Graphs
- Testing results
- Questionnaire results

Example results table

Function	Test Cases	Passed	Failed
Registration	5	5	0
Login	5	5	0
Data submission	5	5	0
Record management	5	5	0
Report generation	5	5	0
Total	**25**	**25**	**0**

7. Discussion

This is where you explain **what the results mean**.

Don't simply repeat your Results section.

For example:

The results demonstrate that the application was able to perform the required functions reliably. The successful completion of all functional test cases indicates that the core features were implemented according to the identified requirements. This finding is consistent with previous studies that reported improvements in efficiency after replacing manual processes with digital systems. However, the evaluation was conducted with a relatively small number of users, which may limit the generalisability of the findings.

You should also discuss:

- Whether your objectives were achieved
- How your results compare with previous research
- Why your results are important
- Strengths of your application
- Limitations

- Unexpected findings

8. Conclusion

The conclusion should answer:

“What did this study accomplish?”

For example:

This study developed and evaluated a mobile application for student attendance management. The application successfully implemented the required attendance recording, management, and reporting functions. Functional testing demonstrated that the core features operated as intended, while user evaluation indicated that the application was easy to use. The study demonstrates the potential of a digital attendance system to reduce the limitations associated with manual attendance management. Future work could focus on integrating additional authentication methods, expanding the system to multiple institutions, and conducting evaluation with a larger user population.